EMERGING TECHNOLOGIES

Emerging Technology Diffusion Agent Record

Jianbo Sun | MASY1-GC 1800 | Assignment 5 | 25 October 2026

Candidate v0.2 finds broad but uneven U.S. workplace use of generative AI. The diffusion stage for retail reporting remains unknown. A fast-follower retailer can investigate now, with verification and workflow complexity as the leading case-specific concern. Diffusion informs timing; the Adoption specialist must decide adoption.

A. Agent and Version

Field	Student record
Student / team	Jianbo Sun
Assignment / specialist agent	Assignment 5 / Emerging Technology Diffusion Agent v0.2
Date	25 October 2026
GitHub repository	https://github.com/Sun417-jianbo/special-dark Target repository; Assignment 5 files not yet published.
Branch	assignment-5-diffusion-jianbo-20261025 Remote branch exists at existing main; public upload pending approval.
Commit	11146059452eb07ad54b673bfcbc41539b790912 Local source commit, preserved in the included Git bundle.
Frozen scaffold version	MASY1800_ET_Agent_Scaffold_v1_0

B. Specialist Purpose and Context

Field	Student record
Governing analytical question	How far has the technology spread in a defined population, what enables or inhibits use, and how should context change investigation timing?
Emerging technology	Generative-AI language-model systems drafting and summarizing business text; broad uptake is distinct from this application.
Intended application	Internal weekly sales narratives drafted from verified SQL/spreadsheet tables with human review.
Organization / industry	Hypothetical U.S. online apparel retailer, 120 employees. Peer population: U.S. online apparel retailers with 50-249 employees.
Adoption posture	Moderate / fast-follower. Two clean contrasts change only posture to aggressive / leading-edge and conservative / risk-averse.
Risk / consequence context	Existing manual templates and one reviewer; no approved production AI data route. No measured staff acceptance, accuracy or time savings. Synthetic or approved non-sensitive tables can support learning.
Management question / time horizon	Next three months, evidence cutoff 25 October 2026. No company deployment or pilot occurred.

C. Three-Level Finding

Field	Student record
ET generally	Broad but uneven U.S. workplace use beyond a niche; exact Rogers category withheld. Q3 2026 RPS estimates: 40.55% of employed adults 18-64 report job use; 34.84% report work use on at least one day last week (S1-S2). Neither measures cumulative adoption or retention.
ET for this application	Unknown for the defined retail peer population and weekly-reporting task. Worker percentages cannot establish firm or application prevalence. The organization being pre-adoption does not establish that peers are early adopters.
ET for this application in this organization	Fast-follower organization is pre-adoption. Begin peer learning and design a bounded investigation. Verification and workflow complexity is the leading likely inhibitor, inferred from the case rather than measured staff perceptions.
Bounded management implication	Aggressive posture prioritizes sandbox learning sooner and peer references in parallel. Conservative posture seeks stronger peer/workflow evidence before a trial proposal. Facts, confidence and safeguards stay stable. The Adoption specialist decides whether to proceed.
Confidence / uncertainty	Moderate confidence in the scoped U.S. worker signal; low confidence in exact peer stage or local repeat use. Quarterly estimates lack uncertainty intervals in this packet; a decline does not prove saturation, reversal or a causal mechanism.
What evidence or condition would change the conclusion	Collect comparable peer task use and repeated local participation. Proposed four weekly reporting cycles track attempted/completed reports, repeat participation, abandonment, errors and total drafting/review/correction effort. This is a diagnostic proposal, not an executed test or a validated threshold.
Rogers factor	**Case assessment and evidence needed**
Relative advantage	Unproven locally. Compare accuracy and total drafting, review and correction effort with the existing manual process.
Compatibility	Potential fit with weekly templates; production data route is unresolved. Check analyst and reviewer workflow fit.
Complexity	Leading inferred inhibitor: reliable verification and workflow integration. Measure correction effort and user difficulties.
Trialability	Bounded learning is feasible with synthetic or approved non-sensitive tables and human review. This does not authorize production use.
Observability	Peer task results are missing. Show comparable, dated before/after reports with denominator and quality checks; attendance or registrations do not prove repeated use.

D. Evidence and Testing

Field	Student record
Most important evidence/source(s)	S1-S2 were updated 20 October 2026. Q3 rates fell 4.65 and 4.39 percentage points from Q2 but remain above Q3 2025. They share one RPS source. S3 broad business-AI use has a different unit, question and window; S4 supplies regional qualitative barriers. S0 supports the framework.
Primary test - result	Baseline plus four revised outputs passed the supplied validator. Four inputs and five outputs passed checks of the schema keywords used by the contracts. All 32 original scaffold files remain byte-identical. These checks do not validate analytical truth.
Contrast test - what appropriately stayed stable	Mechanical input comparisons confirm that two contrasts change only adoption_posture; the scope challenge only appends an unverified claim. Manual same-assistant review found stable evidence, broad-stage caution, peer uncertainty and confidence across revised cases.
Contrast test - what appropriately changed	Posture changes sequencing and priority of learning, not truth or evidence standards. The challenge rejects worker-to-firm extrapolation, an automatic 16% tipping rule, registration counts as active adoption and an immediate company rollout.
Weakness/failure found	v0.1 already used the latest quarter and withheld a precise category, but listed generic factors and vague timing. It did not prioritize a bottleneck, specify measurable follow-up or clearly distinguish change agents from opinion leaders. Its files are preserved.
Revision made and why	v0.2 adds denominator/time discipline, evidence status and tests for each factor, a prioritized inhibitor, repeated-use measures, actor distinctions and bounded posture-sensitive timing. The adversarial claim is a test input, not a factual source.
Remaining limitation / abstention condition	Same-session model-authored outputs and review; no independent model run, independent judge or organizational pilot. Peer-stage uncertainty remains. More relevant peer observations and independent repetitions would strengthen the handoff.

E. Independent Judgment and Handoff

Field	Student record
Is the agent credible enough for team consideration? Why?	Proposed judgment for my review: bring v0.2 as a testable candidate. Broad uptake supports investigation, but does not settle the retail task stage. Verification/workflow complexity matters most in this case. A fast-follower should learn now, with any adoption decision reserved for the next specialist.
Most important recommendation to the team	Strongest signal: recent, scoped U.S. worker uptake. Inhibitor: review/workflow burden. A trainer may act as change agent; a respected reviewer is an opinion leader only if peers seek and trust that advice. Verify those roles. Transfer peer uncertainty, task measures, data/training/reviewer dependencies and timing conditions to Adoption. No verified standard or network effect is assumed.
AI / verification note	ChatGPT/Codex drafted instructions, synthetic cases, saved responses and this report; inspected original-source locations and ran local scripts. Five saved outputs came from this conversation, not an external inference API. Student source review, independent judgment and personal reflection remain to be confirmed.

F. Self-Reflection

AI-assisted draft for my review and personalization before submission. It does not assert independent execution or source verification.

The main distinction I want to carry forward is between growing attention and actual use. This assignment gives me a more precise question to ask: who is using the technology, for which task, and how often? A percentage about workers using AI cannot tell me how many clothing retailers use it for weekly reports. Even a recent-use measure does not show whether the same people continue using it over time.

The quarterly figures also make me more cautious about describing diffusion as a smooth upward curve. The newest point is below the previous quarter, but that alone does not tell me why it fell or whether the decline will continue. I would describe the evidence as broad but uneven use and leave the exact application stage open until there is a relevant peer sample.

The most useful change to the agent is making the five factors lead to specific questions. For this reporting example, I would care more about checking the output and fitting it into the existing workflow than how easy it is to type a prompt. An aggressive company might investigate sooner, while a conservative company might collect stronger peer evidence first. Neither preference changes the underlying facts. Before using this agent with the team, I need to review its source limits and be able to explain why diffusion evidence informs timing without deciding adoption by itself.

Source References

S0. Rogers, E. M. (1983). Diffusion of Innovations, 3rd ed. Chapter 1 (attributes and actors); Chapter 7, pp. 245-247 (idealized categories and incomplete adoption). Author text inspected in a hosted scan; no current uptake claim. https://teddykw2.wordpress.com/wp-content/uploads/2012/07/everett-m-rogers-diffusion-of-innovations.pdf

S1. Bick, Blandin and Deming. Real-Time Population Survey, work-use series RPSGENAIUSAGESHAREWORK, FRED, updated 20 October 2026. Q3 2026: 40.54550%; Q2: 45.19932%. U.S. employed adults 18-64, self-report, not seasonally adjusted. https://fred.stlouisfed.org/series/RPSGENAIUSAGESHAREWORK

S2. Same authors and survey, recent-work-use series RPSGENAIUSAGESHARELWWORK, updated 20 October 2026. Q3: 34.84421%; Q2: 39.23370%. At least one workday in the last week; not an independent dataset or cohort-retention measure. https://fred.stlouisfed.org/series/RPSGENAIUSAGESHARELWWORK

S3. Grundy, Breaux and Khatiwoda (26 May 2026). Large Firms With at Least 20 Employees Biggest AI Users. U.S. Census Bureau. Historical window 14 December 2025-3 May 2026; broad AI, not the retail GenAI task. https://www.census.gov/library/stories/2026/05/ai-use-businesses.html

S4. Gutkowski, Gascon and Silvanus (16 July 2026). New Survey Findings on AI Adoption and Its Effects on Employment and Productivity. St. Louis Fed. May 2026 regional business-contact survey; qualitative mechanisms, not national retail prevalence. https://www.stlouisfed.org/on-the-economy/2026/jul/new-survey-findings-ai-adoption-eighth-district

C0. Supplied Assignment 5 memo and course reading, The Diffusion of Emerging Technologies: Patterns, Actors, and Strategic Market Adoption (date unstated). Course framing; a stylized category boundary is not a guaranteed market tipping point. Full reading not redistributed.

Sources checked by Codex on 25 October 2026. Claim boundaries, exact observations and source locations are preserved in evidence/source_register.json and rps_observations.csv.